Soil Capital for Crop Production: Law of the Minimum

Ryan McWay† Nicholas Gallagher

† *Applied Economics, University of Minnesota*

NatCap TEEMs
University of Minnesota
March 23rd, 2026

Agriculture's Challenges

- Farmers face two major challenges
- A necessity to sequester carbon and methane emissions amassing one third of global emissions contributing to climate change. (Vermeulen et al., 2012; Crippa et al., 2021)
  - While we are making transitions in the energy sector and transportation, limited change in agriculture.
- Modern field management has depleted the macro-nutrients for soil health resulting in soil degradation and soil erosion (Lal, 2015; Louwagie et al., 2009; Montogmery, 2007; Jie et al., 2002)
  - Over extraction of the land leaves farmers more vulnerable to climatic shocks (i.e., the American Dust Bowl).

Soil Capital: Natural Capital vs. Agrochemicals

- We conjecture that soil capital, the natural capital of the soil, is a key input for crop production and a potential substitute for agrochemical inputs.
- Soil capital is a stock of nutrients, organic matter, and microbial life that supports plant growth and ecosystem services.
- Agrochemicals are synthetic inputs such as fertilizers, pesticides, and herbicides that enhance crop growth and protect against pests and diseases.
  - While agrochemicals can boost yields in the short term, they can also degrade soil health and reduce long-term productivity.
  - Nitrogen fertilizer is the most widely used agrochemical input for crop production, but it also contributes to greenhouse gas emissions and water pollution.
- The law of the minimum states that crop yield is limited by the scarcest input, which may be soil capital or agrochemicals. (Brisson et al., 2010)
  - If soil capital is the limiting factor, then farmers can increase yields by investing in soil health rather than increasing agrochemical inputs.

Research Questions

Research Questions

1. Can we estimate the shifts in crop yields using soil capital as an input?
   - Does it follow a CES production function or law of the minimum specification?
2. What is the tradeoff between soil capital and agrochemical inputs?

Data

- UW Forage and Soil Lab's WIDATCP certified labs soil summaries
  - County-year panel for Wisconsin (1995–2017)
  - Mean organic matter (OM, %)
- USDA NASS county corn grain yield (bu/acre) (1997 - 2017)
- Nitrogen fertilizer (1987 - 2017)

Production Functions

- There are two reasonable specifications we may expect to model agricultural production.

1. Constant Elasticity of Substitution (CES) Production Function:
   - Common in economics literature as a flexible functional form that nests both perfect substitutes and perfect complements.
   - However, it may not capture the nonlinearities in the yield response to inputs, especially if there are threshold effects or diminishing returns.
2. Mitcherlich-Baule (Law of the Minimum) Specification:
   - More consistent with agronomic theory and empirical evidence that crop yield is limited by the scarcest input.
   - However, it may be more difficult to estimate and interpret, especially if there are interactions between inputs or unobserved heterogeneity across farms.

Inputs

- Setting: Corn production in Wisconsin
- Objective: Per Acre Corn Yields
- Key Inputs:
  - Soil Capital: Mean Organic Matter (OM, %)
    - Ideal Level: 3-5%, but many farmers have less than 2% OM. (University of Minnesota Extension, 2023; Laboski and Peters, 2012)
  - Agrochemical Input: Nitrogen fertilizer per planted corn acre
    - Ideal Level: 180 - 250 lbs/acre, but many farmers apply more than 300 lbs/acre. (USDA National Agricultural Statistics Service, 2022; USDA Economic Research Service, 2023)

Wisconsin Data and Outcome

- County-year panel for Wisconsin (1995–2017)
- Outcome: county corn grain yield (bu/acre)
- Soil covariate: mean organic matter (OM, %)
- Fertilizer covariate: nitrogen per planted corn acre
- Final nonlinear sample: 1,415 observations across 65 counties
- Estimation via nonlinear least squares (Levenberg–Marquardt)

CES Specification

- Constant Elasticity of Substitution (CES) Specification:

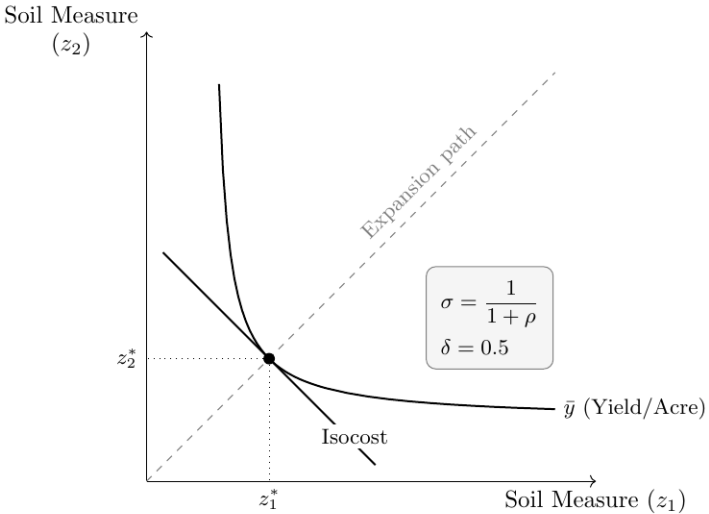
$$y_{ct} = A_{ct}\left[\delta K_{ct}^{-\rho} + (1-\delta)N_{ct}^{-\rho}\right]^{-\frac{1}{\rho}} \tag{1}$$

- OM shifters:

$$A_{ct} = \alpha_0 + \alpha_1 OM_{ct}, \quad K_{ct} = \beta_0 + \beta_1 OM_{ct} \tag{2}$$

- Parameter interpretation:
  - A_{ct}: Hicks-neutral productivity shifter
  - K_{ct}: effective capital/input composite shifted by OM_{ct}
  - N_{ct}: second input in the CES aggregator
  - $\delta \in (0,1)$: share parameter between inputs
  - $\rho = \frac{\sigma-1}{\sigma}$, so $\sigma = \frac{1}{1+\rho}$ is the elasticity of substitution

CES Specification



Law of the Minimum (LOM) Specification

- Mitcherlich-Baule (Liebig's Law of the Minimum) Specification:

$$y_{ct} = \alpha_{ct}\left(1 - \beta_{ct}\exp(-\gamma X)\right) \tag{3}$$

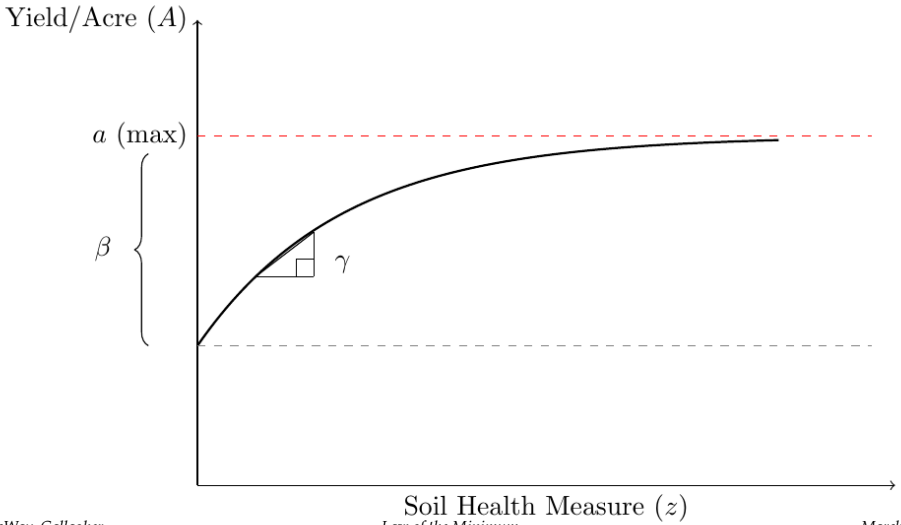
- Specific Specification:

$$y_{ct} = A_{ct}[1 - B_{ct}\,\exp(-C_{ct}\,N_{ct})] \tag{4}$$

$$A_{ct} = \alpha_0 + \alpha_1 OM_{ct}, \quad B_{ct} = \beta_0 + \beta_1 OM_{ct}, \quad C_{ct} = e^{\gamma_0 + \gamma_1 OM_{ct}}$$

- where $X \in \{OM, N\}$ is the input subject to diminishing returns
- α_{ct} is the baseline yield potential, β_{ct} is the plateau parameter, and γ is the decay rate of the yield response to X.
- The law of the minimum implies that yield is limited by the scarcest input, which may be soil capital or agrochemicals.

LOM Specification



CES Parameter Estimates

Table: CES Model Parameter Estimates for Wisconsin Corn Yield

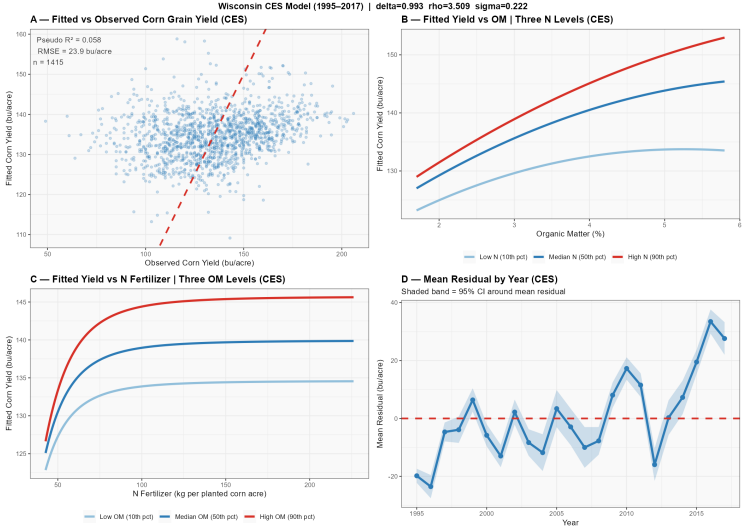
Parameter	Interpretation	Estimate	Std. Error	t-value	p-value	95% CI Low	95% CI High
α_0	Baseline Yield Shifter (bu/acre)	111.17220	25691342.08155	0.000	1.0000	-50354919.30765	50355141.65204
α_1	Yield Shifter Sensitivity to OM	-4.31681	997593.79038	-0.000	1.0000	-1955288.14595	1955279.51233
β_0	Capital Composite Intercept	1.01481	236175.68712	0.000	1.0000	-462903.33194	462905.36156
β_1	Capital Composite Sensitivity to OM	0.13561	31561.29698	0.000	1.0000	-61860.00646	61860.27769
d (logit δ)	Logit Share Parameter (logit δ)	4.95164	816552.60433	0.000	1.0000	-1600438.15285	1600448.05613
ρ	Substitution Parameter	3.50859	1.28815	2.724	0.0065	0.98381	6.03338

Notes: Derived $\delta = 0.9930$ (input share), $\sigma = 0.2218$ (elasticity of substitution). δ recovered via $\delta = 1/(1 + e^{-d})$.

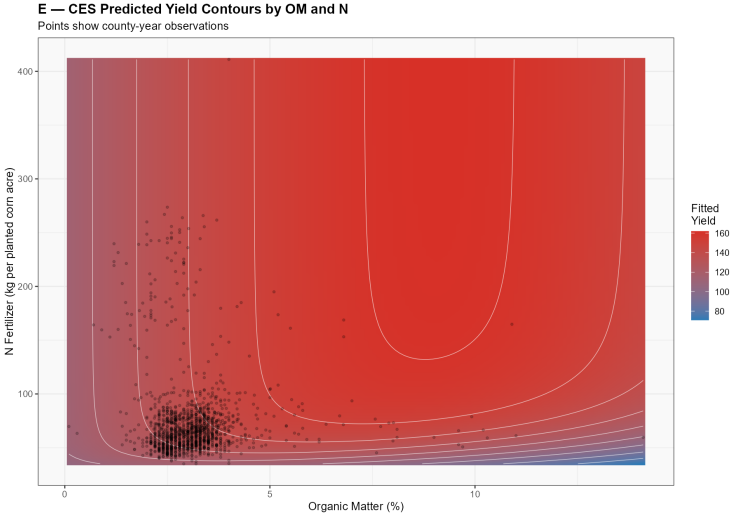
CES Model Fit Summary

- Pseudo-R^2: 0.0576
- RMSE: 23.92
- Residual SE: 23.97
- Derived $\delta = 0.993$ (share weight on K); derived $\sigma = 0.222$ (elasticity of substitution)
- Most parameters are statistically insignificant; model is poorly identified

CES Results Visualized



CES OM–N Tradeoff Contours



LOM Parameter Estimates

Table: Model Parameter Estimates for Wisconsin Corn Yield

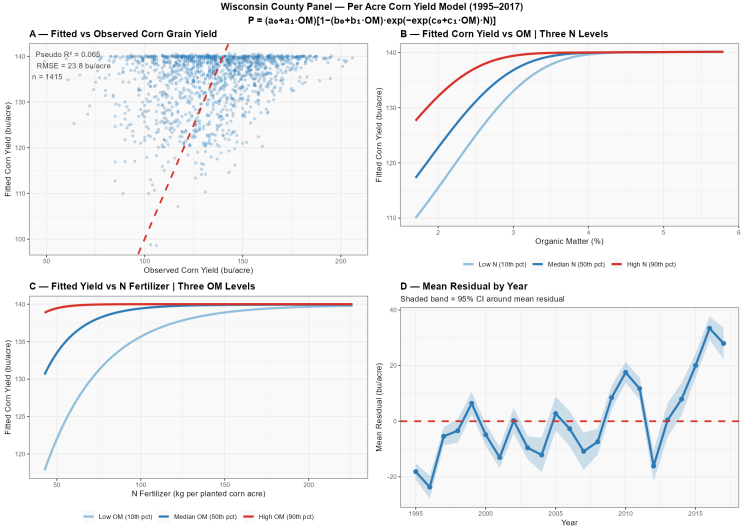
Parameter	Interpretation	Estimate	Std. Error	t-value	p-value	95% CI Low	95% CI High
α_0	Baseline Yield (bu/acre)	139.74030	3.45959	40.392	0.0000	132.95951	146.52109
α_1	Yield increase per 1% OM	0.06846	0.81697	0.084	0.9332	-1.53279	1.66971
β_0	Decay Plateau Parameter	0.40676	0.21534	1.889	0.0591	-0.01530	0.82883
β_1	Decay Plateau Sensitivity to OM	0.05840	0.12644	0.462	0.6442	-0.18942	0.30623
γ_0	Decay Rate (log scale)	-0.79374	0.60224	-1.318	0.1877	-1.97413	0.38665
γ_1	Decay Rate Sensitivity to OM	0.80591	0.18322	4.398	0.0000	0.44679	1.16503

Notes: γ_0 and γ_1 are estimated on the log scale through $\gamma = \exp(\gamma_0 + \gamma_1 \cdot \text{OM})$.

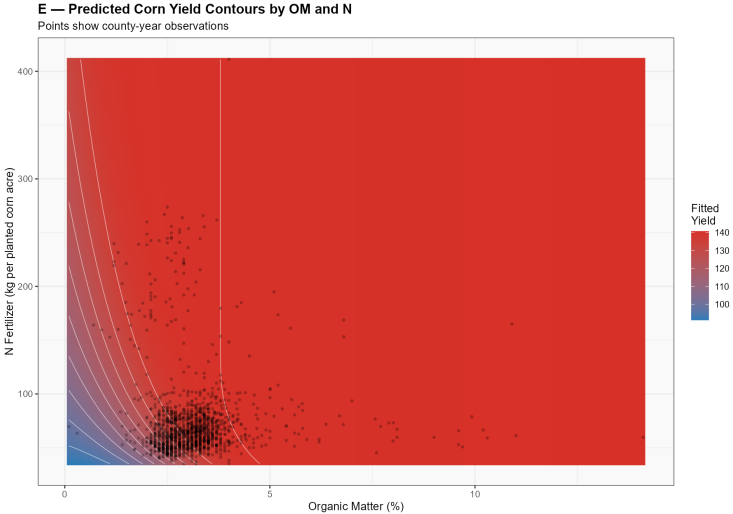
LOM Model Fit Summary

- Pseudo-R^2: 0.0648
- RMSE: 23.83
- Residual SE: 23.88
- Estimated decay sensitivity $\gamma_1 > 0$ and statistically significant

LOM Results Visualized



LOM OM–N Tradeoff Contours



Summary

- Comparison of two production function specifications: CES and LOM
- LOM specification fits the data better and is more consistent with agronomic theory
- Tradeoff contours suggest that soil capital can substitute for agrochemical inputs, but the degree of substitution depends on the levels of both inputs and the yield response to them.
- The results have implications for farmers' decisions on how to invest in soil health and agrochemical inputs, as well as for policymakers' efforts to promote sustainable agriculture and reduce greenhouse gas emissions.

Next Steps

- Explore some other soil macronutrients (e.g., soil pH, Potasium, Phosphorus)
- Explore other crops and regions (e.g., soybeans, national state data)
- Explore influence of regenerative agriculture practices on soil capital and yields (e.g., cover cropping, no till)

The End

Thank You for Your Time!

Contact: *mcway005@umn.edu*
Website: *mcwayrm.github.io*

References I

Brisson, N., P. Gate, D. Gouache, G. Charmet, F.-X. Oury, and F. Huard (2010). Why are wheat yields stagnating in Europe? A comprehensive data analysis for France. *Field Crops Research 119*(1), 201–212.

Crippa, M., E. Solazzo, D. Guizardi, F. Monforti-Ferrario, and F. N. T. an A Leip (2021). Food Systems are responsible for a third of global antropogenic GHG emissions. *Nature Food 2*, 198–209.

Jie, C., C. Jing-zhang, T. Man-zhi, and G. Zi-tong (2002). Soil degradation: A global problem endangering sustainable development. *Journal of Geographical Science 12*, 242–252.

Laboski, C. A. and J. B. Peters (2012). Nutrient application guidelines for field, vegetable, and fruit crops in wisconsin. Technical Report A2809, University of Wisconsin–Extension, Madison, WI.

Lal, R. (2015). Restoring Soil Quality to Mitigate Soil Degradation. *Sustainability 7*, 5875–5895.

Louwagie, G., S. H. Gay, and A. Burrell (2009). Addressing soil degradation in EU agriculture: relevant processes, practices and policies. *JRC Scientific and Technical Reports*.

References II

Montogmery, D. R. (2007). Soil erosion and agricultural sustainability. *Proceedings of the National Academy of the Sciences 104*, 13268–13272.

University of Minnesota Extension (2023). Soil organic matter in cropping systems. https://extension.umn.edu/soil-management-and-health/soil-organic-matter-cropping-systems. Accessed 2024.

USDA Economic Research Service (2023). Fertilizer use and price: Documentation and data sources. Technical report, U.S. Department of Agriculture, Economic Research Service.

USDA National Agricultural Statistics Service (2022, May). Agricultural chemical use survey: Corn, cotton, and rice highlights, 2021 crop year. Technical report, U.S. Department of Agriculture.

Vermeulen, S. J., B. M. Campbell, and J. S. I. Ingram (2012). Climate Change and Food Systems. *Annual Review of Environment and Resources 37*, 195–222.